Matrices (AI HL)

Matrix arithmetic

1 Matrix A has rows $(2, -1)$ and $(3, 4)$; matrix B has rows $(0, 5)$ and $(-2, 1)$. Find the rows of $A + 2B$.

2 For the same matrices A (rows $(2, -1)$, $(3, 4)$) and B (rows $(0, 5)$, $(-2, 1)$), find the rows of the product AB, using row-times-column multiplication.

Determinants and inverses of 2×2 matrices

3 Matrix M has rows $(3, 5)$ and $(2, 4)$. Find $\det(M)$, using $\det(M) = ad - bc$ for rows (a, b) and (c, d), and use it to determine whether M is invertible.

4 Find the inverse of M (rows $(3, 5)$ and $(2, 4)$), using $M^{-1} = \frac{1}{\det(M)}$ times the matrix with rows $(d, -b)$ and $(-c, a)$.

Solving systems of equations using matrices

5 Write the system $3x + 5y = 11$, $2x + 4y = 8$ in matrix form, then use M^{-1} (found in the previous problem, rows $(2, -2.5)$ and $(-1, 1.5)$) to solve for x and y.

Eigenvalues and eigenvectors

6 Matrix M has rows $(4, 1)$ and $(2, 3)$. Find the eigenvalues of M, using $\det(M - \lambda I) = 0$.

7 Find the eigenvector corresponding to $\lambda = 5$ for the matrix M (rows $(4, 1)$ and $(2, 3)$) from the previous problem.

Applying matrices to transition/Markov models

8 A weather model has a transition matrix T with first row $(0.7, 0.4)$ and second row $(0.3, 0.6)$ (columns: today's sunny/rainy state; rows: tomorrow's sunny/rainy probabilities). If today is sunny (state vector with entries 1 then 0), find the probability distribution for tomorrow's weather.

Diagonalization and powers of a matrix

9 Matrix M (rows $(4, 1)$, $(2, 3)$) has eigenvalues $\lambda = 5$ and $\lambda = 2$ (found earlier), with eigenvectors $(1, 1)$ and $(1, -2)$ respectively. Explain, in general terms, why diagonalizing a matrix makes computing large powers like M^{10} much easier than repeated direct multiplication.

Determinants of 3×3 matrices

10 Matrix N has rows $(1, 2, 3)$, $(0, 1, 4)$, and $(5, 6, 0)$. Find $\det(N)$ by cofactor expansion along the first row.

Identity matrix and matrix powers

11 Explain why $MI = IM = M$ for any square matrix M and the identity matrix I of matching size, and use this idea to find M^2 where M has rows $(1, 2)$ and $(0, 1)$.

Using the inverse to solve a real-world system

12 A shop sells two products. Selling 2 units of product A and 3 units of product B brings in 76 dollars. Selling 4 units of A and 1 unit of B brings in 68 dollars. Write this as a matrix equation and solve for the price of each product using the coefficient matrix M with rows $(2, 3)$ and $(4, 1)$.

Long-run behavior of a Markov chain

13 For the weather transition matrix T from earlier in this worksheet (rows $(0.7, 0.4)$ and $(0.3, 0.6)$), find the steady-state probability distribution $(p, 1-p)$, which satisfies $T(p, 1-p)^T = (p, 1-p)^T$.

Singular matrices and when an inverse fails to exist

14 Matrix N has rows $(2, 4)$ and $(1, 2)$. Show that N has no inverse, and explain what this means about the corresponding system of linear equations.

Transpose of a matrix and its properties

15 Matrix A has rows $(1, 2, 3)$ and $(4, 5, 6)$. Write down A^T (the transpose), and state its dimensions.

16 Explain, using the general rule $(AB)^T = B^T A^T$, why the order of multiplication must be reversed when taking the transpose of a matrix product.

17 Matrix A has rows $(2, 0)$ and $(0, 5)$. Find A^2 and A^3, and describe the pattern for A^n in general.